

GIT-

- ppt
- diagrams
- MCQs

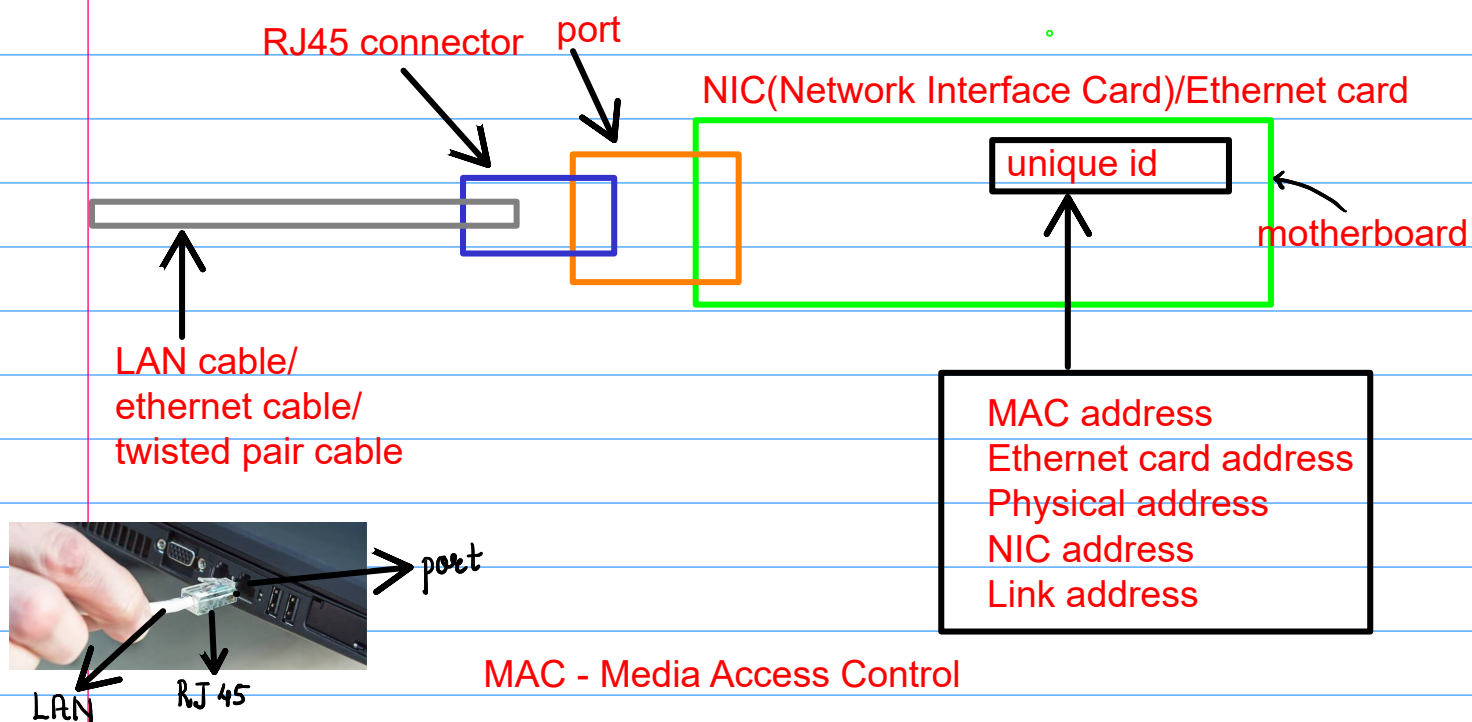
Books-

- P K Sinha
- Forouzan

Quiz-

- 1. DCN_01 } 10Qs
- 2. DCN_02 }
- 3. DCN_ModuleEnd } 20Qs

- hosts/ devices/IP addresses/nodes/machines

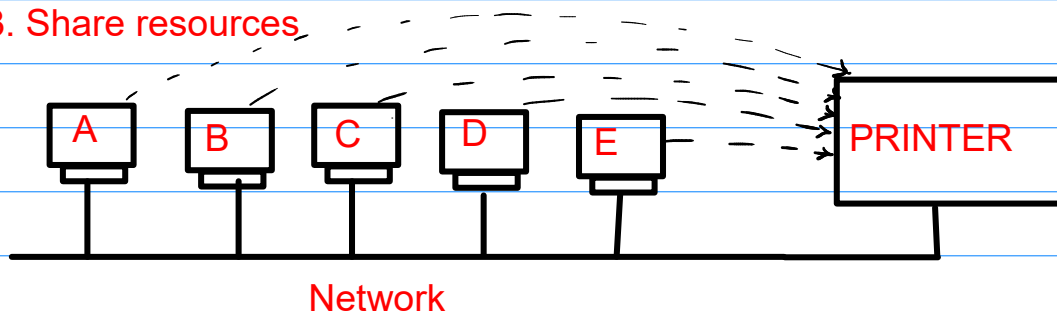


4 fundamental characteristics for effective data communication-

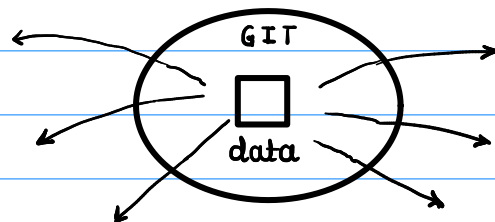
1. Delivery
2. Accuracy
3. Timeliness
4. Jitter - delay

Need of Network-

1. Information sharing
2. Enhanced communication - group
3. Share resources



4. Centralized management - GIT /google drive



5. Remote computing - zoom(remote access)
 - Anydesk
 - team viewer
 - MC remote computing
 - Chrome remote desktop, etc

Network Criteria

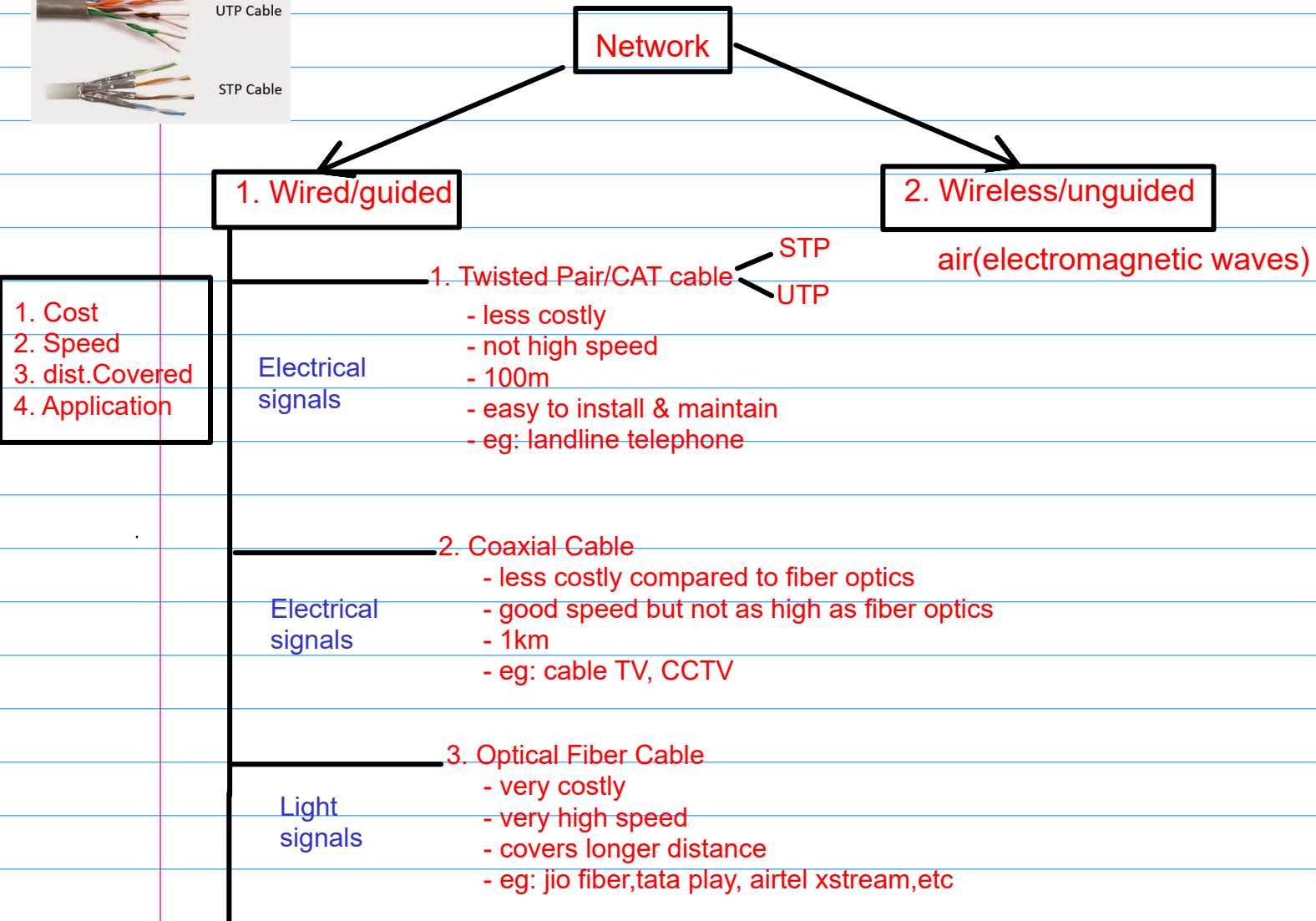
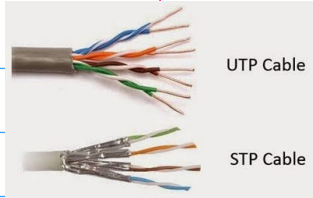
1. Speed
2. cost
3. performance - number of users, type of transmission medium, hardware, software

Twisted
Pair cable coaxial fiber
optic

- delay & throughput(actual rate of data transfer across the n/w in given time(bits/sec))
4. Reliability - robust, freq. of failure should be least
 5. Security - https

STP - Shielded Twisted Pair

UTP - Unshielded Twisted Pair



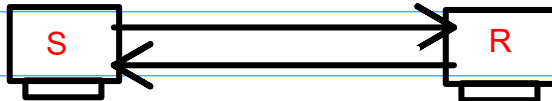
Transmission Mode /Data Flow Direction

1. Simplex mode



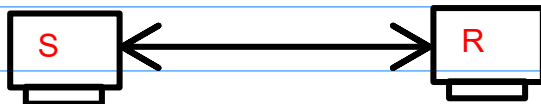
- Data transfers only in one direction
- eg: radio, keyboard, TV remote, pager, etc

2. Half Duplex Mode



- Data transfers in both directions but not at the same time (not simultaneously)
- eg: walkie talkie, FAX, etc

3. Full Duplex Mode

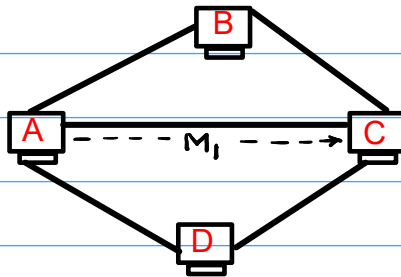


- Data transfers in both directions simultaneously
- eg: telephone, mobile, chats, voice call, video call, etc

SWITCHING

- Selection of best route to transmit data across the network is called as Switching

1. Circuit Switching

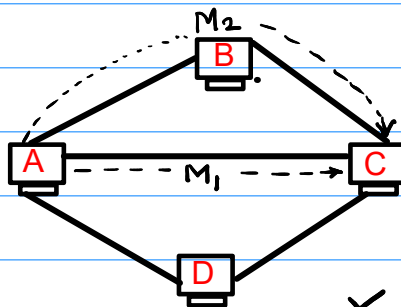


C A M1

C - destination
A - source
M1 - message

1. Connection established
 2. Data transferred
 3. Connection is terminated
- eg: telephone

2. Message Switching



- store and forward technique
- eg: SMS

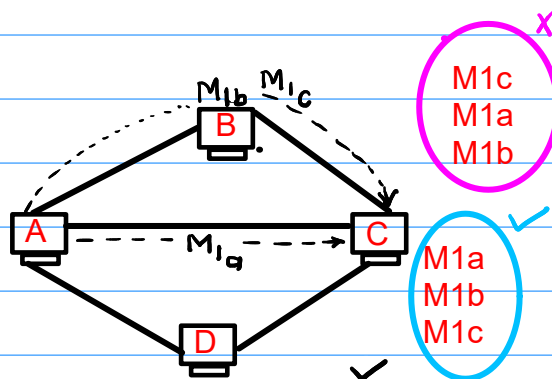
C A M1

✓
C A M2

3. Packet Switching

ethernet technology

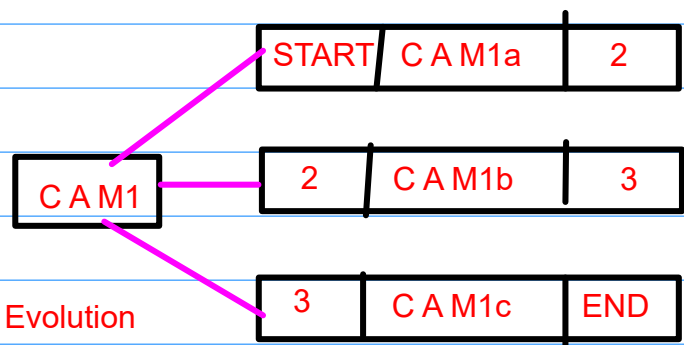
1 data packet = 1500 bytes



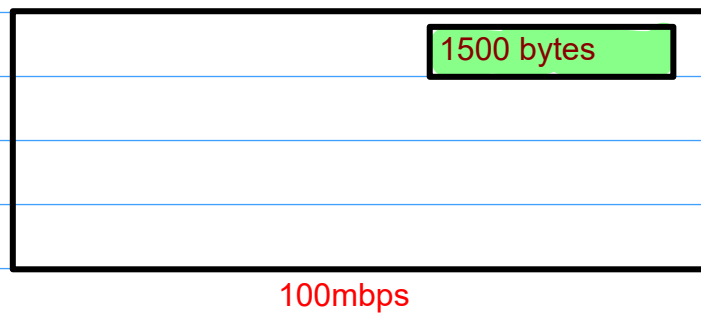
M1 = 4000 bytes

M1a = 1500 bytes
M1b = 1500 bytes
M1c = 1000 bytes

eg: VoLTE - Voice Over Long Term Evolution



Multiplexing



Bandwidth-

- capacity of wired or wireless network (medium) to transfer maximum data through the network in a given period of time.

to overcome this wastage of bandwidth we use

MULTIPLEXING

FDM

TDM

multiplexing - many to one (mux)
demultiplexing - one to many (demux)

FDM - Frequency Division multiplexing

TDM - Time Division Multiplexing

FDM(frequency Division Multiplexing)

radio FM

- 91.1 ✓
- 93.5
- 94.2
- 92.7
- 104.2
- 105
- 91.9
- 98.3

